BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: 122101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Which is the heighest melting halide.

(A) Nacl
(B) Na Br
(C) Na F
(D) None of these

Q2. The transition metals copper and _____ are best electrical conductors.

(A) Iron
(B) Vanadium
(C) Chromium
(D) Manganese

Q3. Oxidation number of "Pt" in $K_2[PtCl_6]$ is _____.

(A) 1
(B) 2
(C) 4
(D) 6

Q4. Petrochemicals are used in the manufacture of.

(A) Polythene
(B) Detergents
(C) Fibers
(D) All of these

Q5. The number of Hydrogen in Pyrole is/are _____.

(A) 5
(B) 9
(C) 4
(D) 3

Q6. The general formula of Alkenes having two C=C bonds is _____.

(A) C_nH_{2n+2}
(B) C_nH_{2n}
(C) C_nH_{2n-1}
(D) C_nH_{2n-2}

Q7. How many isomers are possible for C_2H_6O.

(A) 2
(B) 3
(C) 8
(D) 5

Q8. Which one posses greater number of carbons?

(A) Diesel Oil
(B) Kerosine Oil
(C) Gasoline
(D) Pet Ether

Q9. The optimum P^H of pepsin is.

(A) 4.5
(B) 1
(C) 2.5
(D) 2.0

Q10. Ethanol is produced from starch by the process of.

(A) Hydrolysis
(B) Hydrogenation
(C) Decomposition
(D) Fermentation

Q11. Lactose is a dissacharide of _____.

(A) Glucose & Maltose
(B) Glucose & glactose
(C) Glucose & glucose
(D) Fructose & lactose

Q12. R-CH(X)-R is an example of which alkyle halide?

(A) Primary
(B) Secondary
(C) Tertiary
(D) None of these

Q13. Which one of the following is synthetic polymer?

(A) Cellulose
(B) Protein
(C) Rubber
(D) Polyethene

Q14. Cu^{+2} salt solution is blue in colour due to transition of electron from _____ orbitals.

(A) "S" to "P"
(B) "d" to "d"
(C) "P" to "d"
(D) "P" to "P"

Q15. Which one of the following halide is most reactive towards nucleophilic substitution reaction.

(A) C_2H_5Br
(B) C_2H_5I
(C) C_2H_5F
(D) CH_2H_5cl

Q16. Catalytic hydration of Ethyne gives.

(A) Aldehyde
(B) Ketone
(C) Ester
(D) None of these

Q17. The general formula of cycloalkanes is.

(A) C_nH_{2n+2} (B) C_nH_{2n}

(C) C_nH_{2n-1} (D) C_nH_{2n-2}

Q18. Dehydrohalogenation of alkyle halide carried out in the presence of _____.

(A) Zn duct (B) H_2SO_4

(C) Alcoholic KOH (D) Aqueous KOH

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 40

QII. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) Why Aldehydes and Ketones undergo nucleophilic addition while Alkenes undergo electrophilic addition.

(ii) What is LUCAS Test?

(iii) How and why RCOOH is stronger acid than Alcohols (ROH)?

(iv) How can you prepare Grignard reagent?

(v) How RX are prepared from Alcohols?

(vi) Give any four characteristic of organic compounds.

(vii) Explain melting point trends in IA and IIA group.

(viii) Relate the inert pair effect with the Fajan's rule in explaining the formation of ionic or covalent compounds by the elements of the same group.

(ix) Differentiate between addition and condensation Polymer.

(x) What are pesticides? Draw structure of DDT.

(xi) How coal is converted into petroleum through Fischer-Tropsch process.

(xii) Write down the main function of Infrared (IR) spectroscopy.

(xiii) Write down the formation of PVC and Nylon 6,10.

SECTION "C"

Maximum Marks: 27

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Explain Markowni kor's rule in addition reaction of Alkene with suitable example.

(b) [5] Discuss the orientation effect of ortho/para and meta directing groups for dis substitution in benzene ring.

QIV.

(a) [5] Give IUPAC name of the following. i) CH_3-C≡C-CH(CH_3)$_2$ ii) benzene ring with -CHO and -OH substituents (salicylaldehyde structure) iii) CH_3-CH(O-CH_2-CH_3)-CH_2-CH_3 iv) CH_2(OH)-CH(OH)-CH_2(OH) v) CH_2=CH-COOH

(b) [4] Prove the Resonance stability of Benzene.

QV.

(a) [5] Describe the mechanism of electrophilic substitution reaction in benzene.

(b) [4] Discuss the reaction of Grignard reagent with. i) Aldehyde ii) Ketones iii) Esters iv) Carbon dioxide

QVI.

(a) [5] Discuss the structure of carbon dioxide and silicon dioxide.

(b) [4] Write a note on waste water treatment.